STATIC STRUCTURAL ANALYSIS

Connecting Rod

Finite Element Analysis | Ansys Workbench 2026 R1 | SolidWorks 2026

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Programme: B.Tech Mechanical Engineering — 3rd Year
Software: SolidWorks 2026 | Ansys Workbench 2026 R1 (Student Edition)

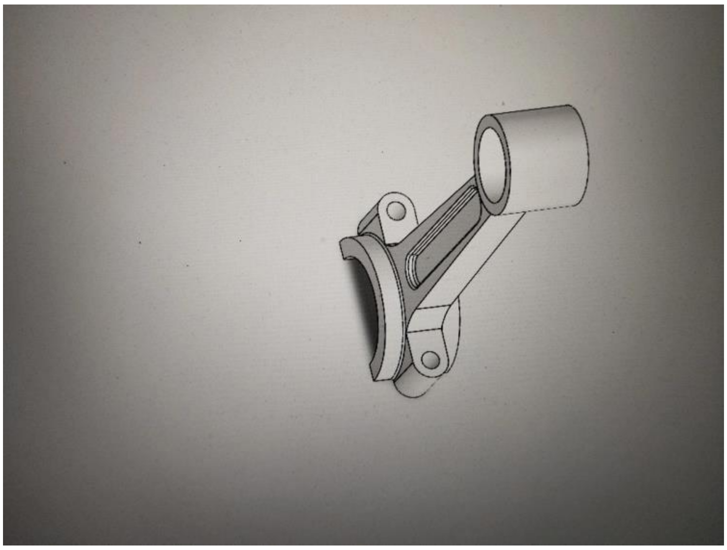


Figure 1 — SolidWorks 3D model of the connecting rod

Abstract

This report presents the static structural finite element analysis (FEA) of a connecting rod designed in SolidWorks 2026 and analysed in Ansys Workbench 2026 R1 (Student Edition). A compressive force of 16,000 N was applied at the small end bore to simulate peak combustion loading in a 4-cylinder petrol engine. The analysis evaluates equivalent von Mises stress, total deformation, and factor of safety. Independent manual calculations for cross-sectional area, second moment of area, radius of gyration, slenderness ratio, and analytical safety factor are presented alongside FEA results. The minimum FEA safety factor obtained is 3.87 at the small end boss fillet, confirming the design is structurally adequate under the applied static load.

1. Introduction

The connecting rod is one of the most critically loaded components in a reciprocating internal combustion engine. It connects the piston pin (small end) to the crankshaft journal (big end), converting the piston's linear reciprocating motion into the rotational motion of the crankshaft. During the power stroke the rod carries high compressive loads from combustion gas pressure acting on the piston crown. During the intake and exhaust strokes it is pulled in tension. This alternating stress cycle makes it a fatigue-critical part in engine design.

This project covers the complete workflow: 3D solid modelling in SolidWorks 2026, STEP export, static structural FEA in Ansys Workbench 2026 R1, and independent manual stress verification. Results confirm the geometry is adequate under the specified loading.

2. Objectives

- Model the connecting rod in SolidWorks 2026 using sketch-based solid modelling features.
- Export as STEP AP214 and import into Ansys Workbench Static Structural.
- Assign Low Alloy Steel 4140 (normalized) from the Ansys Engineering Data library.
- Apply boundary conditions — Fixed Support at big end bore, 16,000 N force at small end bore.
- Obtain von Mises equivalent stress, total deformation, and safety factor contour results.
- Carry out manual stress calculations and compare with FEA output.

3. Component Description and Dimensions

The connecting rod modelled here is representative of a small 4-cylinder petrol engine. The geometry has three main regions: the small end boss (piston pin end), the tapered shank with a central weight-reduction slot, and the big end with a split cap retained by two bolts. For this analysis the assembly is treated as a single monolithic solid body. Key dimensions from the SolidWorks drawing are listed below.

Dimension	Value	Unit
Small end bore diameter (d_s)	Ø 33.80	mm

Small end outer diameter (D_s)	Ø 45.00	mm
Big end bore diameter (d_b)	Ø 72.00	mm
Centre-to-centre shank length (L)	90.00	mm
Shank slot fillet radius (r)	R 8.00	mm
Big end cap bolt hole diameter	Ø 9.00	mm
Small end outer radius	R 22.50	mm
Overall assembly width	76.00	mm

Figure 2 — SolidWorks rendered view of the connecting rod model

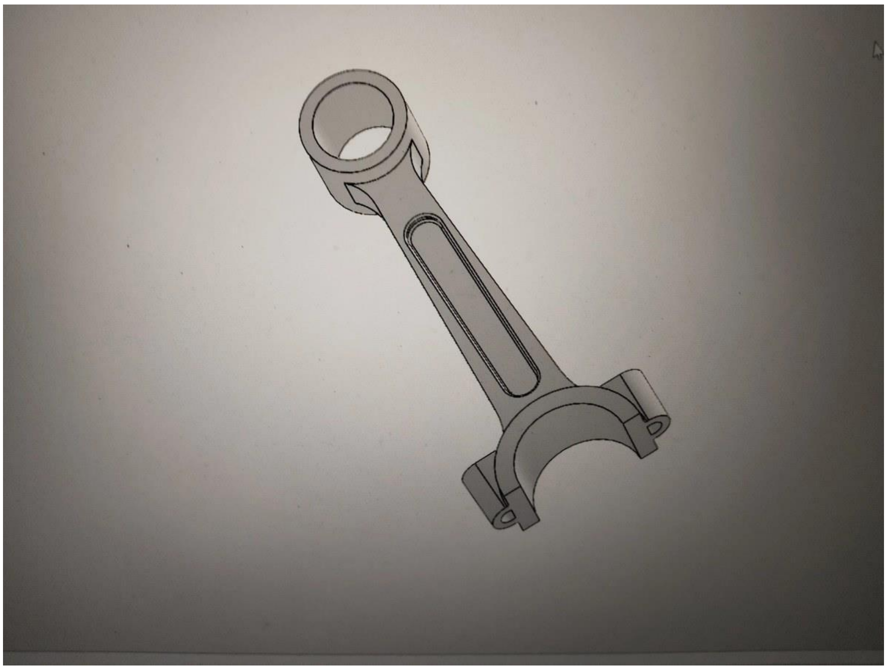


Figure 3 — SolidWorks model, isometric view showing full rod geometry

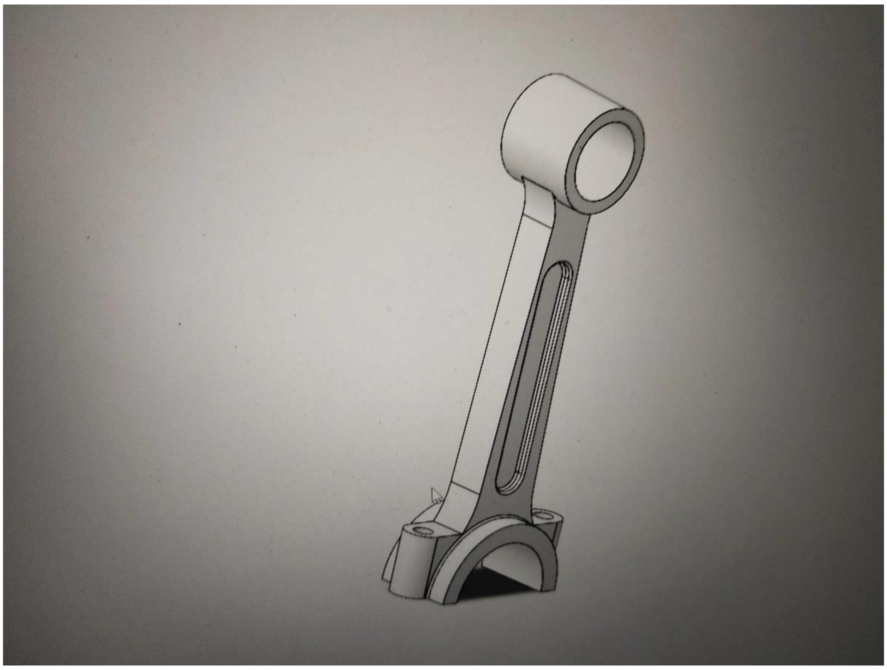


Figure 4 — SolidWorks model, big end and shank detail

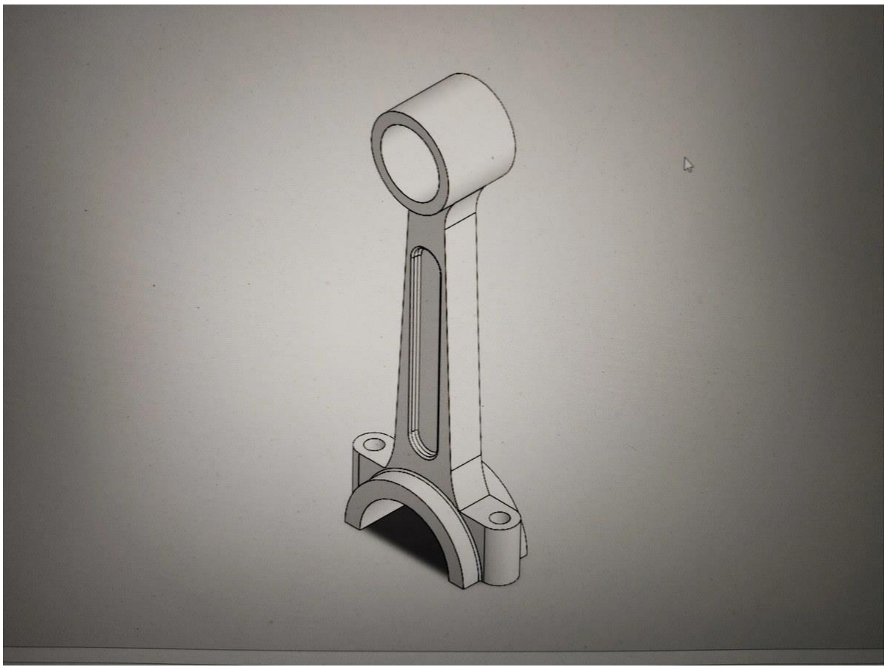


Figure 5 — SolidWorks model, big end cap view showing bolt lugs

4. Material Properties

Low Alloy Steel 4140, normalized condition, was assigned from the Ansys Engineering Data library. This alloy is widely used for connecting rods in medium-duty engines because of its high tensile strength, good fatigue resistance, and machinability. Properties used in the analysis are listed below.

Property	Value	Unit
Young's Modulus (E)	2.125×10^{11}	Pa

Poisson's Ratio (v)	0.29	—
Tensile Yield Strength (Sy)	652.2	MPa
Tensile Ultimate Strength (Su)	1015	MPa
Bulk Modulus	1.686×10^{11}	Pa
Shear Modulus	8.236×10^{10}	Pa
Thermal Conductivity	43.33	W/m·°C
Density	7850	kg/m³

5. Manual Stress Calculations

Manual analytical calculations were done independently to estimate the direct axial stress, second moment of area, radius of gyration, slenderness ratio, and factor of safety at the critical section — the small end boss. These are compared with FEA in Section 8. Original handwritten sheets are in Appendix A.

5.1 Given Data

Quantity	Value
Applied Force (F)	16,000 N (compressive, axial)
Small end bore diameter (d_s)	33.80 mm
Small end outer diameter (D_s)	45.00 mm
Shank length, centre-to-centre (L)	90.00 mm
Fillet radius (r)	8.00 mm
Yield Strength (S_y)	652.2 MPa
Young's Modulus (E)	2.125×10^5 MPa

5.2 Cross-Sectional Area at Small End

Critical section is the annular cross-section at the small end boss. For a hollow circle:

$$A = (\pi/4) \times (D_s^2 - d_s^2)$$
$$A = (\pi/4) \times (45^2 - 33.8^2)$$
$$A = (\pi/4) \times 882.56$$
$$A = 693 \text{ mm}^2$$

5.3 Second Moment of Area and Radius of Gyration

$$I = (\pi/64) \times (D_s^4 - d_s^4)$$
$$I = (\pi/64) \times (4{,}100{,}625 - 1{,}305{,}765)$$

$$I = 137{,}220 \text{ mm}^4$$
$$k = \sqrt{(I/A)} = \sqrt{(137{,}220/693)} = 14.07 \text{ mm}$$

5.4 Axial Compressive Stress

$$\sigma_{axial} = F / A = 16{,}000 / 693 = 23.09 \text{ MPa}$$

Since σ_{axial} = 23.09 MPa << S_y = 652.2 MPa → No yielding under direct axial load.

5.5 Slenderness Ratio and Buckling Check

$$L_{eff} = 0.5 \times L = 45 \text{ mm} \quad \text{(both ends pinned)}$$
$$\lambda = L_{eff} / k = 45 / 14.07 = 3.20$$

λ = 3.20 << 40 → Short column. Euler buckling does not govern; direct stress controls.

5.6 Stress Concentration at Fillet

$$K_t \approx 1.5 \quad \text{(fillet, } r / d = 8 / 33.8 = 0.24\text{)}$$
$$\sigma_{peak} = K_t \times \sigma_{axial} = 1.5 \times 23.09 = 34.64 \text{ MPa}$$

FEA captures full 3D stress concentration including bending. The FEA peak of 168.55 MPa is more accurate — the difference reflects combined bending and notch effects that 1D hand calculation cannot model.

5.7 Factor of Safety

$$FoS_{direct} = S_y / \sigma_{axial} = 652.2 / 23.09 = 28.25$$
$$FoS_{corrected} = S_y / \sigma_{peak} = 652.2 / 34.64 = 18.83$$
$$FoS_{FEA} = S_y / \sigma_{VM,max} = 652.2 / 168.55 = \mathbf{3.87} \quad \leftarrow \text{governing}$$

The FEA value of 3.87 is the design-governing figure as it accounts for the full 3D geometry and actual stress concentration effects.

6. FEA Setup and Boundary Conditions

6.1 CAD Modelling — SolidWorks 2026

The connecting rod was modelled from scratch in SolidWorks 2026. Sequence: base boss-extrude for the shank body, boss features for small end and big end, cut-extrude for the central weight-reduction slot, big end cap split with bolt lug bosses, fillets applied last. The model was exported as STEP AP214 for Ansys import.

6.2 Material Assignment

Low Alloy Steel 4140 (normalized) was assigned from the Ansys Engineering Data library to all bodies. E = 2.125×10^{11} Pa, ν = 0.29.

6.3 Mesh

Ansys automatic mesher was used with default settings. Higher element density is applied automatically at curved bore surfaces and fillets. No manual sizing was used in this study.

6.4 Boundary Conditions

- Fixed Support — applied to the full inner surface of the big end bore. All 6 DOF constrained, simulating the rod seated on the crankshaft journal.
- Force — 16,000 N at the small end bore inner surface. X = 0, Y = +16,000 N, Z = 0. Simulates peak combustion gas load from the piston, putting the shank in compression.

6.5 Results Requested

- Equivalent (von Mises) Stress
- Total Deformation
- Safety Factor — von Mises criterion, referenced to yield strength

7. FEA Results

7.1 Equivalent (von Mises) Stress

Peak von Mises stress of 168.55 MPa is at the small end boss fillet — the shank-to-boss transition where section change and bending are highest. The big end, being fully fixed, shows near-zero stress. The gradient across the rod is smooth with no unexpected high-stress regions.

Parameter	Value	Unit
Maximum von Mises Stress	168.55	MPa
Minimum von Mises Stress	0.041	MPa
Average von Mises Stress	32.70	MPa
Yield Strength (Sy)	652.2	MPa
Max Stress as % of Yield	25.8 %	—

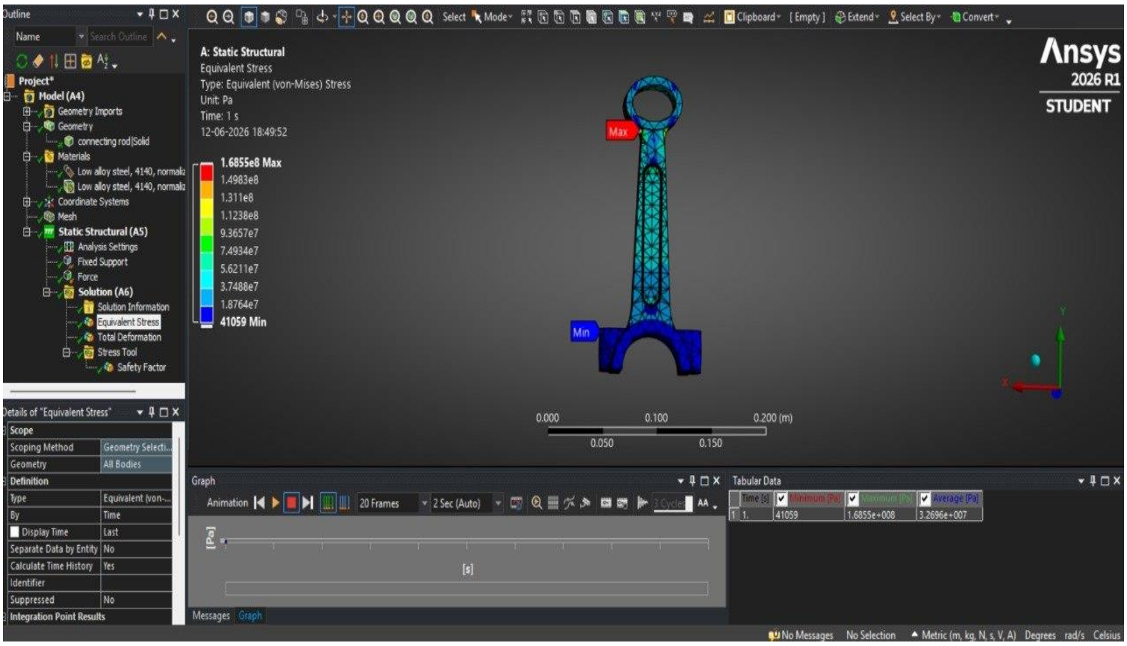


Figure 6 — Equivalent (von Mises) Stress. Max 168.55 MPa at small end boss fillet. Ansys 2026 R1.

7.2 Total Deformation

Maximum deformation of 0.0634 mm occurs at the small end bore — the free end where the load is applied. This is consistent with the fixed constraint at the big end. The smooth gradient from small end to big end confirms elastic behaviour throughout with no spurious displacement zones.

Parameter	Value	Unit
Maximum Total Deformation	6.3419×10^{-5}	m
Maximum (converted)	0.0634	mm
Average Deformation	1.0513×10^{-5}	m
Minimum Deformation	0	m

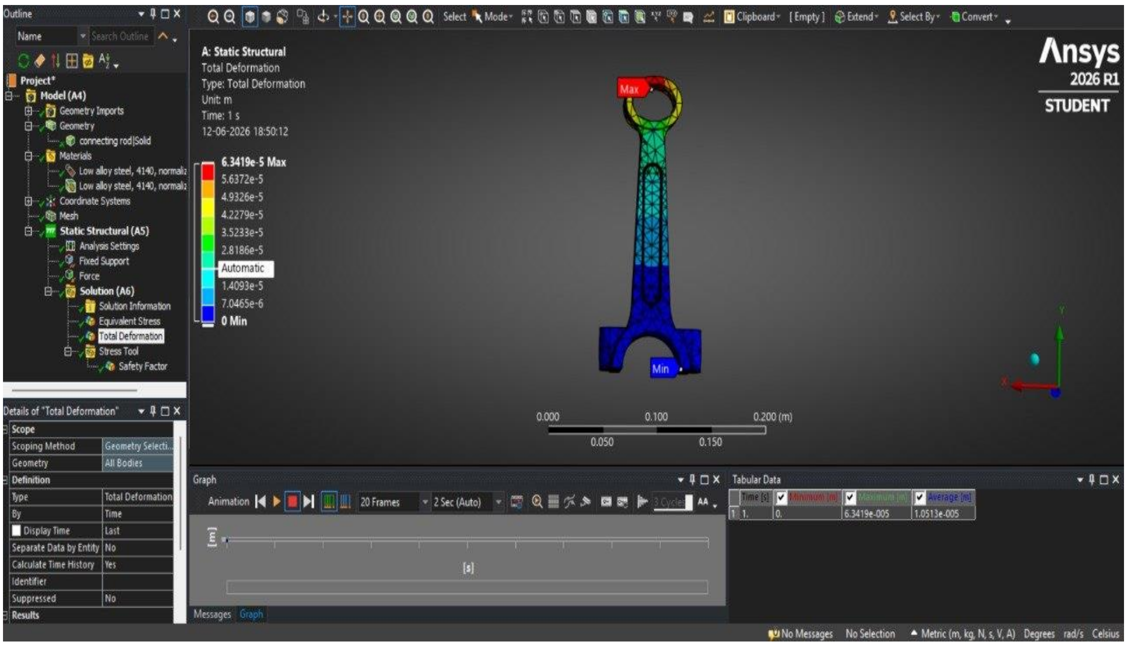


Figure 7 — Total Deformation. Max 0.0634 mm at small end bore. Ansys 2026 R1.

7.3 Safety Factor

Safety Factor was computed using von Mises criterion against yield strength. Ansys display is capped at 15. Minimum FoS of 3.87 appears at the small end boss fillet. The average of 14.198 shows the bulk of the rod is well within the safe zone.

Parameter	Value	Unit
Minimum Safety Factor	3.8695	—
Maximum FoS (display cap)	15.0	—
Average Safety Factor	14.198	—
Criterion	von Mises vs. Yield	—
Location of minimum FoS	Small end boss fillet	—

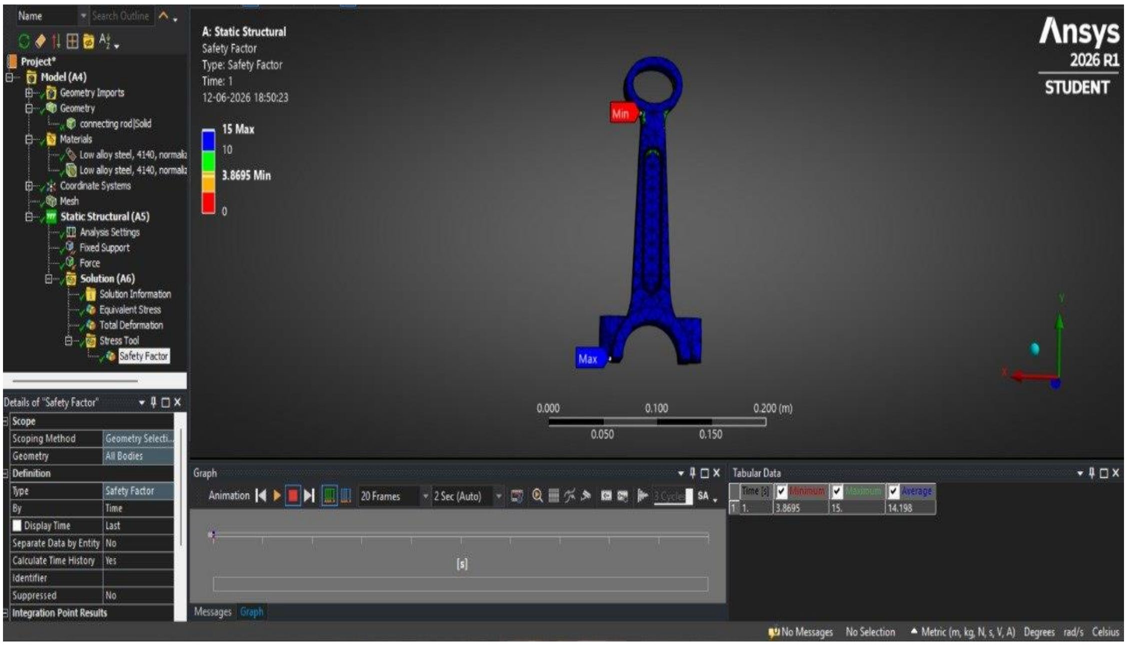


Figure 8 — Safety Factor. Minimum FoS = 3.87 at small end. Ansys 2026 R1.

8. Comparison: Manual Calculations vs FEA

The table below compares key results from the analytical hand calculations with Ansys FEA output. Both methods use the same geometry and loading but differ in assumptions and level of detail.

Parameter	Manual Calculation	Ansys FEA
Cross-section area (A)	693 mm²	From geometry
2nd moment of area (I)	137,220 mm⁴	—
Radius of gyration (k)	14.07 mm	—

Slenderness ratio (λ)	3.20 (short column)	—
Max stress	23.09 MPa (axial)	168.55 MPa (VM)
Kt-corrected stress	34.64 MPa	Included in FEA
FoS (direct)	28.25	—
FoS (Kt corrected)	18.83	—
FoS (governing)	18.83	3.87 ← design value
Max deformation	Not computed	0.0634 mm
Failure prediction	No yielding ✓	No yielding ✓

The difference between manual FoS (18.83) and FEA FoS (3.87) is expected. Hand calculation treats the load as purely axial on a full annular section with a single Kt factor. FEA captures the 3D geometry, combined axial and bending stresses, and the true notch shape at the fillet. Both confirm no yielding; FEA is the governing value.

9. Discussion

The peak stress at the small end boss fillet is expected — this is where the section transitions abruptly and bending moment is highest. The R8 fillet already reduces stress concentration vs a sharp corner; a larger fillet radius would lower peak stress without much weight penalty.

This is a single static load case. In service the rod sees fully reversed loading — compression on power stroke, tension on intake/exhaust. A proper fatigue assessment using S-N data for 4140 normalized steel would be needed to determine endurance life. The static FoS of 3.87 does not directly represent a fatigue safety factor.

Fixing the full big end bore surface is a simplification. In practice the crankshaft bearing carries load through a hydrodynamic film over roughly 180° of the bore. A partial contact model would give a more realistic load path, but the fixed-bore approach is standard for preliminary static analysis.

A mesh convergence study — reducing element size at the small end fillet and monitoring peak von Mises stress — would confirm the sensitivity of the 168.55 MPa result to mesh density. The smooth contour gradients in Figure 6 suggest the result is not grossly mesh-dependent, but a formal check is recommended.

10. Conclusion

- Connecting rod modelled in SolidWorks 2026, analysed under 16,000 N static compressive load in Ansys Workbench 2026 R1.
- Maximum von Mises stress: 168.55 MPa at the small end boss fillet — 25.8% of yield strength. No yielding predicted.
- Maximum total deformation: 0.0634 mm at the piston pin bore — negligible relative to component dimensions.

- Minimum FoS from FEA: 3.87 — within the accepted range of 2.5–4.0 for connecting rod design.
- Manual calculations confirm no yielding: axial stress 23.09 MPa, Kt-corrected 34.64 MPa — both well below yield.
- FEA FoS of 3.87 governs because it captures 3D stress concentration effects that the hand calculation cannot.
- Design is structurally adequate under the modelled static loading conditions.

11. References

1. Shigley, J.E., Mischke, C.R. & Budynas, R.G. (2020). Mechanical Engineering Design, 11th ed. McGraw-Hill.

2. Norton, R.L. (2014). Machine Design: An Integrated Approach, 5th ed. Pearson.

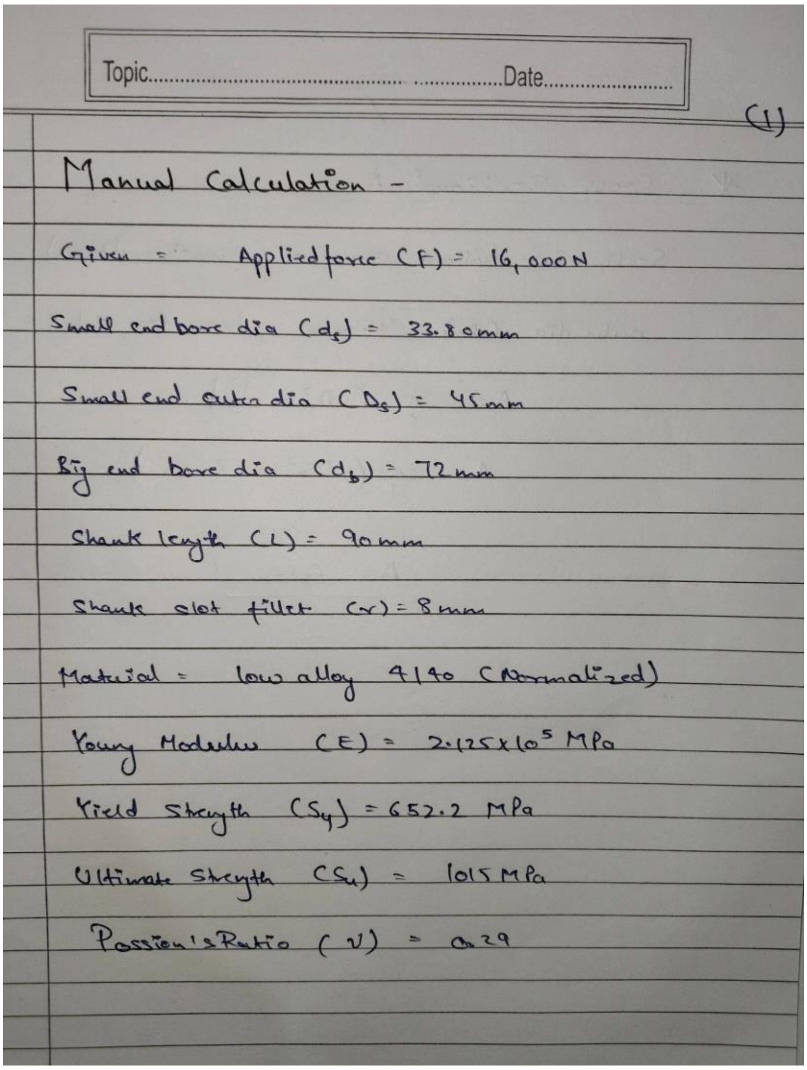
3. Ansys Inc. (2026). Ansys Mechanical User Guide, Release 2026 R1. Canonsburg, PA.

4. Dassault Systemes (2026). SolidWorks 2026 Help Documentation. Vélizy-Villacoublay.

5. Repgen, G. (1998). Optimised Connecting Rods to Enable Higher Engine Performance. SAE Technical Paper 980882.

Appendix A — Handwritten Calculation Sheets

Original handwritten calculation sheets for this project. Sheets 1–8 correspond to the sequence in Section 5.



Topic.................................... ....Date........................

(1)

Manual Calculation -

Given = Applied force (F) = 16,000 N

Small end bore dia (d_s) = 33.80 mm

Small end outer dia (D_s) = 45 mm

Big end bore dia (d_b) = 72 mm

Shank length (L) = 90 mm

Shank slot fillet (r) = 8 mm

Material = low alloy 4140 (Normalized)

Young Modulus (E) = 2.125×10^5 MPa

Yield Strength (S_y) = 652.2 MPa

Ultimate Strength (S_u) = 1015 MPa

Possion's Ratio (ν) = 0.29

Sheet 1 — Given Data and Design Parameters

Topic..Date.......................

(2)

* Cross - Sectional Area

Section :- hollow annular (Small end boss)

outer dia (D_s) = 45 mm , Inner dia (d_s) = 33.80 mm

$$A = \frac{\pi}{4}\left(D_s^2 - d_s^2\right)$$

$$A = \frac{\pi}{4}\left(45^2 - 33.80^2\right)$$

$$A = 693\,mm^2$$

Sheet 2 — Cross-Sectional Area at Small End

Topic..Date......................

(3)

*. Second Moment of area & Radius of Gyration

$$I = \frac{\pi}{64}\left(D_s^4 - d_s^4\right)$$

$$= \frac{\pi}{64}\left((45)^4 - (33.80)^4\right)$$

$$= \frac{\pi}{64}\left(2794860\right)$$

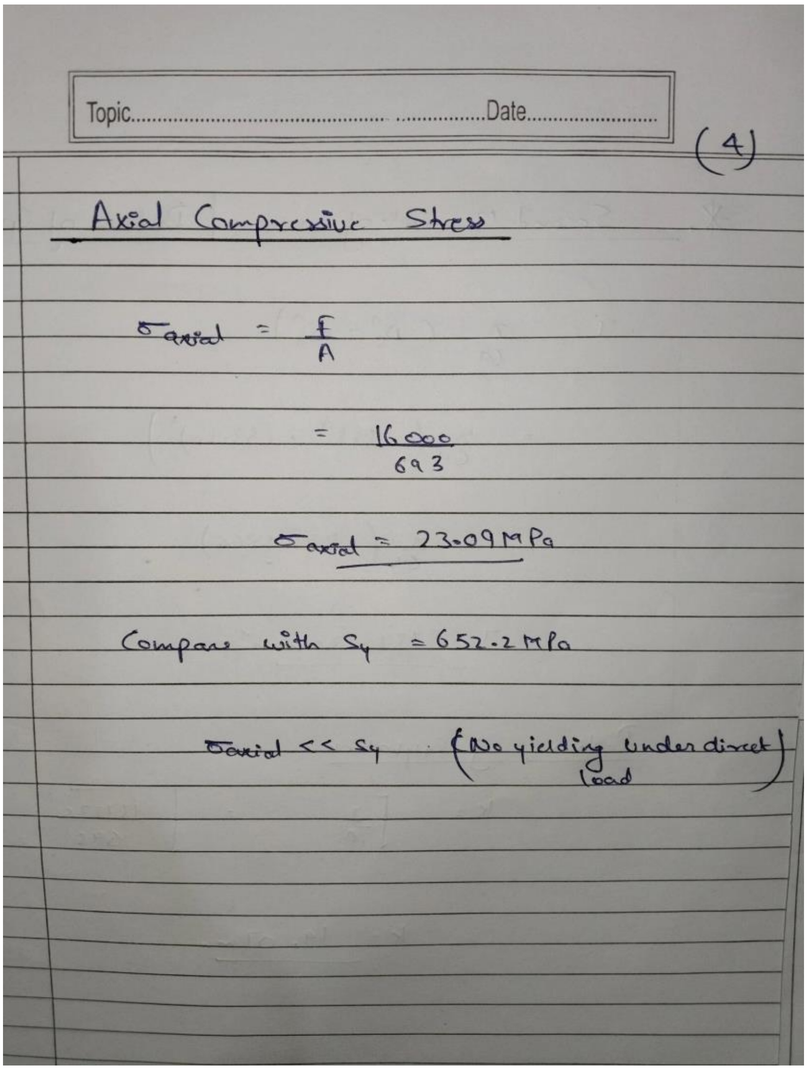
$$\underline{I = 137,220\, mm^4}$$

Radius of Gyration

$$K = \sqrt{\frac{I}{A}} = \sqrt{\frac{137220}{695}}$$

$$\underline{K = 14.07\, mm}$$

Sheet 3 — Second Moment of Area and Radius of Gyration



Topic.. ....Date........................

(4)

Axial Compressive Stress

$$\sigma_{axial} = \frac{F}{A}$$

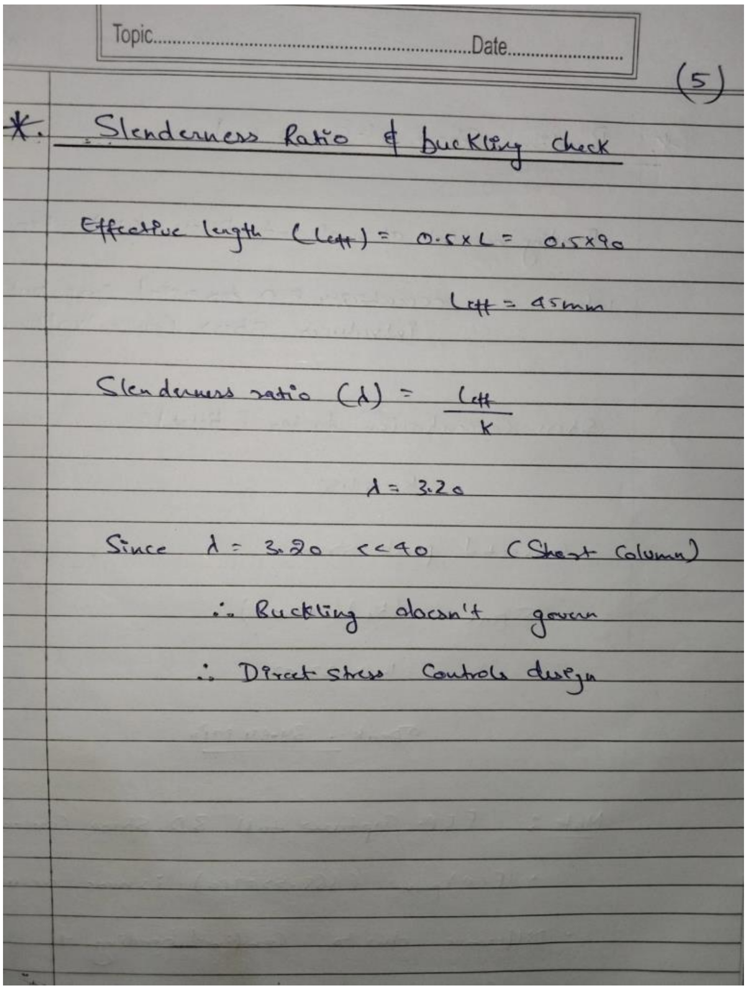
$$= \frac{16000}{693}$$

$$\sigma_{axial} = 23.09\ MPa$$

Compare with $S_y = 652.2\ MPa$

$\sigma_{axial} << S_y$ (No yielding under direct load)

Sheet 4 — Axial Compressive Stress



Topic.......... Date..........

(5)

*. Slenderness Ratio & buckling check

Effective length (L_{eff}) = $0.5 \times L$ = 0.5×90

$L_{eff} = 45$ mm

Slenderness ratio (λ) = $\frac{L_{eff}}{k}$

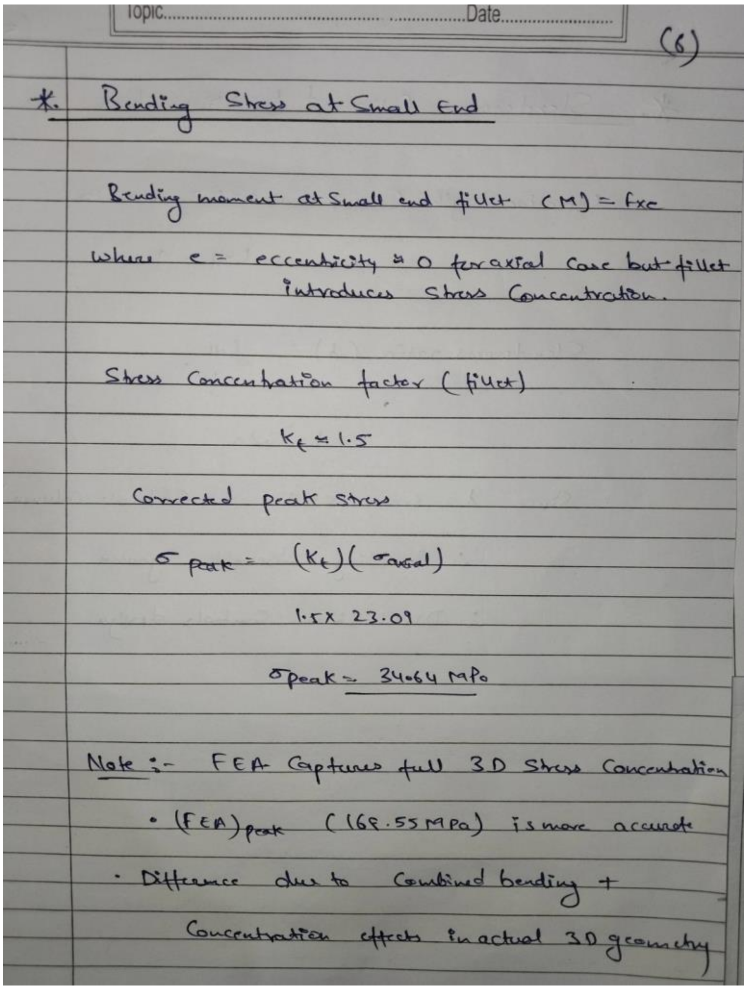
$\lambda = 3.20$

Since $\lambda = 3.20 << 40$ (Short column)

∴ Buckling doesn't govern

∴ Direct stress controls design

Sheet 5 — Slenderness Ratio and Buckling Check



Topic.................................... ....Date.......................

(6)

*. Bending Stress at Small End

Bending moment at Small end fillet $(M) = F \times e$

where e = eccentricity ≈ 0 for axial case but fillet introduces Stress Concentration.

Stress Concentration factor (fillet)

$$K_t \approx 1.5$$

Corrected peak stress

$$\sigma_{peak} = (K_t)(\sigma_{axial})$$

$$1.5 \times 23.09$$

$$\sigma_{peak} = 34.64\ MPa$$

Note :- FEA Captures full 3D Stress Concentration

- $(FEA)_{peak}$ (168.55 MPa) is more accurate
- Difference due to Combined bending + Concentration effects in actual 3D geometry

Sheet 6 — Bending Stress at Small End Fillet

Topic.. Date.........................

(7)

* Factor of Safety

Analytical FoS (direct stress only)

$$(FoS)_{direct} = \frac{S_y}{\sigma_{axial}} = \frac{652.2}{23.09}$$

$$(FoS)_{direct} = 28.25$$

Analytical FoS (with K_t correction)

$$(FoS)_{corrected} = \frac{S_y}{\sigma_{peak}}$$

$$= \frac{652.2}{34.64}$$

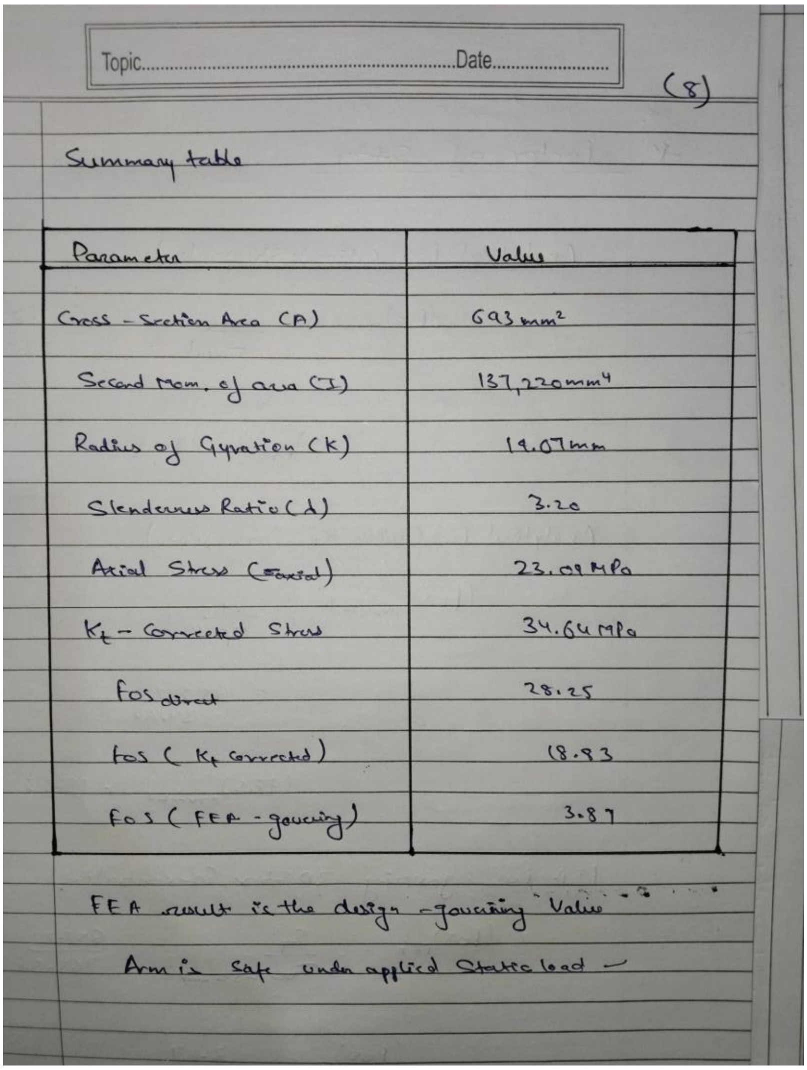
$$(FoS)_{corrected} = 18.83$$

FEA-FoS (governing 3D stress concentration).

$$(FoS)_{FEA} = \frac{S_y}{\sigma_{von\ mises\ max}} = \frac{652.2}{168.55}$$

$$(FoS)_{FEA} = 3.87$$

Sheet 7 — Factor of Safety Calculations



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(8)

Summary table

Parameter	Value
Cross - Section Area (A)	693 mm^2
Second Mom. of area (I)	137,220 mm^4
Radius of Gyration (K)	14.07 mm
Slenderness Ratio (λ)	3.20
Axial Stress (σ_{axial})	23.09 MPa
K_t - Corrected Stress	34.64 MPa
FoS_{direct}	28.25
FoS (K_t corrected)	18.93
FoS (FEA - governing)	3.87

FEA result is the design-governing value.

Arm is safe under applied static load.

Sheet 8 — Summary Table